Post-Implementation Review, Security Validation, and Root Cause Analysis (RCA)

Project Identifier: NVIDIA-REG-RD-2026-1106
Target Environment: Regional Hub — Brussels, Belgium
Lead Systems Integrator: Technical Support & Engineering Taskforce
Documentation State: Final Baseline Engineering Record

1. System Architecture Blueprint

This section outlines the finalized logical and physical design states for the corporate facility network layer. Following comprehensive architectural staging and multi-zone diagnostic verification, the network has been transitioned to a fully routed, stable, stateful, and deterministic environment.

1.1 Completed Departmental VLSM Matrix

Zone Designation	VLAN ID	Network Range	Subnet Mask Size	Default Gateway IP	Host Scope Parameters
Production	VLAN 10	192.168.10.0	255.255.255.0 (/24)	192.168.10.1	Dynamic IP Allocations (.2 to .254)
Support_B	VLAN 20	192.168.20.0	255.255.255.224 (/27)	192.168.20.1	Dynamic IP Allocations (.2 to .30)
Study	VLAN 30	192.168.30.0	255.255.255.224 (/27)	192.168.30.1	Dynamic IP Allocations (.2 to .30)
Support_A	VLAN 40	192.168.40.0	255.255.255.224 (/27)	192.168.40.1	Dynamic IP Allocations (.2 to .30)
Management	VLAN 50	192.168.50.0	255.255.255.224 (/27)	192.168.50.1	Dynamic IP Allocations (.2 to .30)
IT Department	VLAN 60	192.168.60.0	255.255.255.240 (/28)	192.168.60.1	Dynamic IP Allocations (.2 to .14)
Server Pool & Storage SAN	VLAN 70	192.168.70.0	255.255.255.240 (/28)	192.168.70.1	DNS (.10), iSCSI Storage Target (.6)
Hardened Printer Loop	VLAN 80	192.168.80.0	255.255.255.240 (/28)	192.168.80.1	4x Multi-Function Printers (.2 to .5)

2. Chronological Chrono-Log of Engineering Troubleshooting & Interventions

During deployment and commissioning phases, systematic physical and logical path failures were isolated and remediated. Below is the technical Root Cause Analysis (RCA) and resolution ledger for all issues encountered.

2.1 Incident 1: Physical Layer Dropouts (Support_A Offline)

- **Symptom:** Workstations and endpoints inside the Support_A sector could not communicate locally or reach the network core.
- **Root Cause:** Physical layer disconnects. The physical straight-through cabling layout on the access switch (Switch1) for endpoints `PC6` through `PC9` and the local printing appliance was missing or unmapped.
- **Remediation & Resolution:** Terminated physical straight-through Cat6A lines from endpoints to access ports `FastEthernet0/6 - 9` and `0/12`. Forced administrative status to active (`no shutdown`).

2.2 Incident 2: Subnet Mask Discrepancies & SVI Gateway Mismatches

- **Symptom:** Workstations obtained IP addresses via DHCP but completely failed to ping across zone bounds or communicate with the DMZ Web Portal (`192.168.90.2`).
- **Root Cause:** Subnet mask misalignment between the DHCP server scopes and the Core Switch Virtual Interfaces (SVIs). The pools issued narrow `/27` masks (`255.255.255.224`), whereas the switch interfaces had mismatched default allocations, forcing the endpoints to point to an unreachable local gateway address.
- **Remediation & Resolution:** Synchronized all switch gateways to match the DHCP scope structure. Reconfigured the Core-MLS CLI as follows:

```
interface Vlan20
 ip address 192.168.20.1 255.255.255.224
interface Vlan30
 ip address 192.168.30.1 255.255.255.224
interface Vlan40
 ip address 192.168.40.1 255.255.255.224
interface Vlan50
 ip address 192.168.50.1 255.255.255.224
```

2.3 Incident 3: Asymmetric Routing & Firewall Boundary Drops

- **Symptom:** Internal subnets successfully hit the DMZ Server, but packet traces showed an instant timeout on the return path.
- **Root Cause:** Asymmetric stateful routing on the perimeter Cisco ASA Firewall. The firewall's routing engine had no backward pointers for individual `/24` and `/27` subnets, causing its stateful packet inspection engine to drop return packets coming from the `dmz` or `inside` links.
- **Remediation & Resolution:** Implemented a broad internal aggregate summary route on the firewall to encapsulate all internal subnets cleanly under a single next-hop pointer pointing back to the Core switch:

```
route inside 192.168.0.0 255.255.0.0 192.168.80.1
same-security-traffic permit intra-interface
same-security-traffic permit inter-interface
```

2.4 Incident 4: FTP Storage Server Connection Timeouts

- **Symptom:** Attempting to run `ftp 192.168.70.3` returned an immediate session timeout error.
- **Root Cause:** Dual-layer configuration error. First, the server was executing on a completely different zone (DMZ network at `192.168.90.3`), rendering the target `.70.3` address nonexistent. Second, the ASA firewall lacked active application inspection protocols for FTP's secondary dynamic data port constraints.
- **Remediation & Resolution:** Target connections were redirected to the correct IP address (`192.168.90.3`). Concurrently, stateful FTP application protocol inspection was forced into the global security policy on the firewall:

```
policy-map global_policy
 class inspection_default
  inspect ftp
```

2.5 Incident 5: Peripheral Device Gateway Configuration Failures

- **Symptom:** Multi-function printers across office zones were completely inaccessible from other segments, and running Cisco IOS CLI interface assignments threw an `% Invalid input detected` error.
- **Root Cause:** First, printer devices had manual static gateway mappings pointing to foreign subnets (e.g., the Study printer had its gateway mapped to the Support_B network). Second, engineers were trying to run specific physical port configurations directly inside the switch's Privileged EXEC mode prompt (`Switch#`) rather than stepping into Global Configuration mode.
- **Remediation & Resolution:** Flipped the printers to run dynamic DHCP addressing or re-mapped their static parameters to line up perfectly with their respective local subnets. On the leaf switches, the correct command syntax was executed to open the configurations:

```
Switch# configure terminal
Switch(config)# interface FastEthernet0/12
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
```

3. Hardened Security & Isolation Realignment

To support the modern perimeter parameters specified in the corporate contract while ensuring zero data degradation, two strict extended access control filters are actively deployed across the core multi-layer switch fabric.

3.1 Backend Block-Storage Hardening (NVIDIA_SECURITY_ACL)

Applied outbound on the core storage gateway (VLAN 70), this policy restricts raw block-level iSCSI disk access (`192.168.70.6`) strictly to authorized administrative stations, preventing lateral exploitation paths.

```
CORE-MLS# show access-lists
Extended IP access list NVIDIA_SECURITY_ACL
    10 permit ip 192.168.60.0 0.0.0.15 host 192.168.70.6 (Matches logged)
    20 deny ip 192.168.10.0 0.0.0.255 host 192.168.70.6
    30 deny ip 192.168.20.0 0.0.0.31 host 192.168.70.6
    40 deny ip 192.168.30.0 0.0.0.31 host 192.168.70.6
    50 deny ip 192.168.40.0 0.0.0.31 host 192.168.70.6
    60 deny ip 192.168.50.0 0.0.0.31 host 192.168.70.6
    70 permit ip any any (General Traffic Clear)
```

3.2 Printer Loop Isolation (PRINTER_SECURITY_ACL)

To eliminate high-risk printer pivot points, all corporate multi-function printers have been mapped into a dedicated printing loop subnet via **VLAN 80**. The following isolation filter is applied ingress on the SVI interface to drop any unauthorized lateral internal office snooping or packet sniffing attempts:

```
ip access-list extended PRINTER_SECURITY_ACL
  10 permit ip 192.168.60.0 0.0.0.15 192.168.80.0 0.0.0.15
  20 deny ip any 192.168.80.0 0.0.0.15
  30 permit ip any any
exit
```

```
interface Vlan80
 ip access-group PRINTER_SECURITY_ACL in
exit
```

4. Diagnostic Sign-Off Verification Metrics

The total environment has been fully cleared against all verification test parameters:

Verification Sign-Off Matrix:
✅ **Name Resolution Test:** `nslookup stratum.nvidia` successfully queries the server at `192.168.70.10` and resolves to the DMZ Web Server IP address.
✅ **Web Landing Page Access:** Workstations cleanly render the custom landing page via HTTP using the corporate domain name link.
✅ **Active FTP Session States:** Administrators can establish clean sessions to the DMZ FTP Server (`192.168.90.3`) through the firewall.
✅ **iSCSI Protection:** Attempts to contact the raw storage block from the production subnets are successfully dropped by the access switch fabric, while IT terminals retain clean, low-latency communication paths.